

Q1. Please help me find a good electronics circuit simulator freely available on the Internet.

Prashant
Through e-mail

A1. Hundreds of good-quality circuit simulators are freely available on the Internet. If you are new to circuit simulation, choose one that is simple and easy to learn. Otherwise, your interest will die by the time you learn all the complex features and become ready for your first circuit simulation.

Simulation modeling allows designers and engineers to avoid repeated building of multiple physical prototypes to analyse designs for new or existing parts. Before creating the physical prototype, users can virtually investigate many digital prototypes. Electronic circuit simulation uses mathematical models to replicate the behaviour of an actual electronic device or circuit. Simulation software allows modeling of circuit operation and is an invaluable analysis tool.

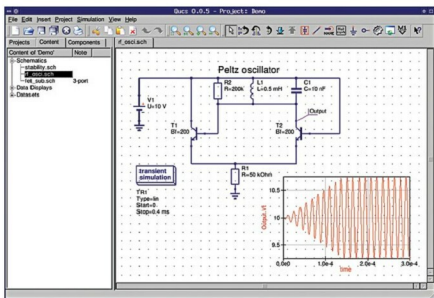
While there are mostly analogue electronics circuit simulators, popular simulators often include both analogue and digital simulation capabilities. This means that any simulation may contain components that are analogue, digital or a combination of both. An entire mixed-signal analysis can be driven from one integrated schematic. So it is best to select a mixed-signal simulation tool.

At EFY Lab, we use gEDA Project. Within the gEDA Project tool chain there are two analogue circuit simulators: ngspice and gnucap.

ngspice is a clean-up of classical SPICE 3f5 for the Linux platform. It is fully functional, and includes XSpice extensions and CIDER framework.

gnucap is a new circuit simulator written from the ground up. It offers the ability to perform event-driven as well as continuous-time simulations.

But if you are new to circuit simulation, I would recommend Qucs. Qucs is an integrated circuit simulator, which means you are able to set up a circuit with a graphical user interface (GUI) and simulate the large-signal, small-signal and noise behaviour of the



Qucs circuit simulator

circuit. After simulation has finished, you can view simulation results on a presentation page or window. Qucs is easy to learn and very user-friendly compared to other tools.

Q2. There are lots of antistatic bags available in the market. How do they differ from one another?

Yashoda

Through e-mail

A2. The capabilities of different kinds of bags are explained below:

Pink poly bags (dissipative poly bags). These bags dissipate a static charge to ground that keeps charge from building up on the package. The material is also antistatic, suggesting that it will not charge up by tribocharge. The material's resistivity is in dissipative range and is usually about 10^9 to 10^{11} . Unfortunately, these bags have no shielding ability. A static field or discharge occurring outside the bag will easily penetrate the bag and damage the electronics inside.

Black conductive poly bags. Black poly bags are highly conductive with surface resistivity of about 10^3 to 10^4 . These dissipate the charge very quickly but, unfortunately, this fast dissipation also creates the possibility of spark on their surface. As the material is conductive, it does provide some

small measure of shielding. However, there is no plastic layer (dielectric) to isolate the device inside the bag. The charge may be transferred through the volume of the material to the device instead of around the material to ground.

Shielding bags. Shield bags have both the dissipative and antistatic attributes of black poly bags but add a metal shield and polyester dielectric to stop static from entering the bag. Shield bags generally stop 97 per cent of a 1000V static pulse, applied to the outside of the bag, from reaching the inside. Pink poly bags stop only about 10 per cent and black poly bags about 30 per cent.

Moisture barrier bags. Moisture barrier bags have dissipation, antistatic and static shielding properties together with moisture vapour barrier. The moisture barrier protects moisture-sensitive items and improves long-term storage.

Then there are other products like pink bubble bags, antistatic foam and antistatic boxes, which are generally used in packaging for improving the reliability of the equipment in the field.

Answer compiled by EFY technical editor, Ankit Gupta. Letters and questions for publication may be addressed to Editor, Electronics For You, D-87/1, Okhla Industrial Area, Phase 1, New Delhi 110020 (E-mail: editsec@efyindia.com) and should include name and address of the sender